Day 1. Introduction of ML

Machine learning is the techinique to let the computer learn how to do something by itself.

What is Machine Learning

> the field of study that gives computers the ability to learn without being explicitly programming

the more oppurtunity you give a learning algorithm to learn, the better it will do?

> why?

Machine learning algorithms

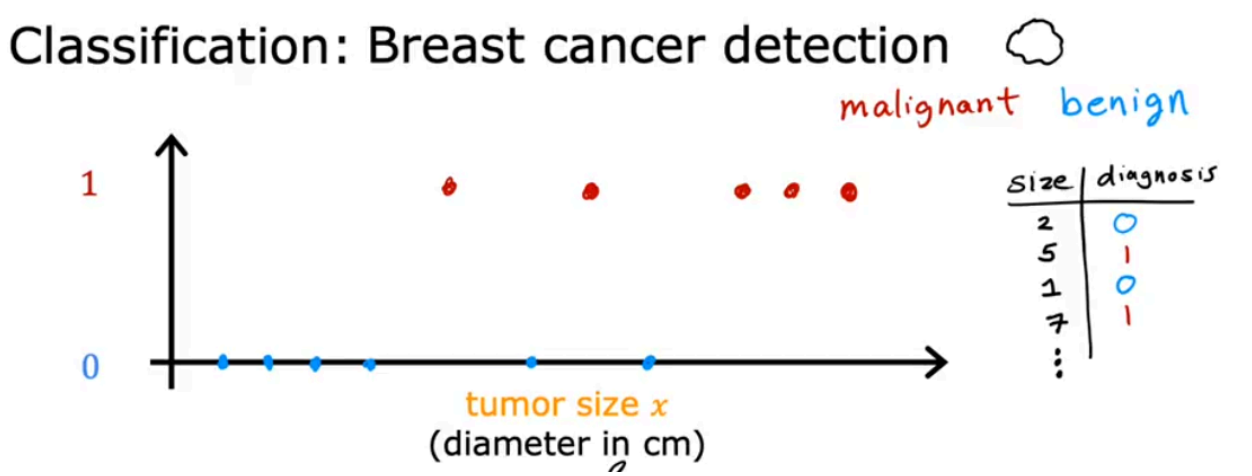
- SUpervised learning
- Unsupervised learning
  - Recommender systems
  - Reinforcement learning
    "we will also cover practical advice for applying learning algorithms, learn best practices of using the right algorithms."

Supervised learning

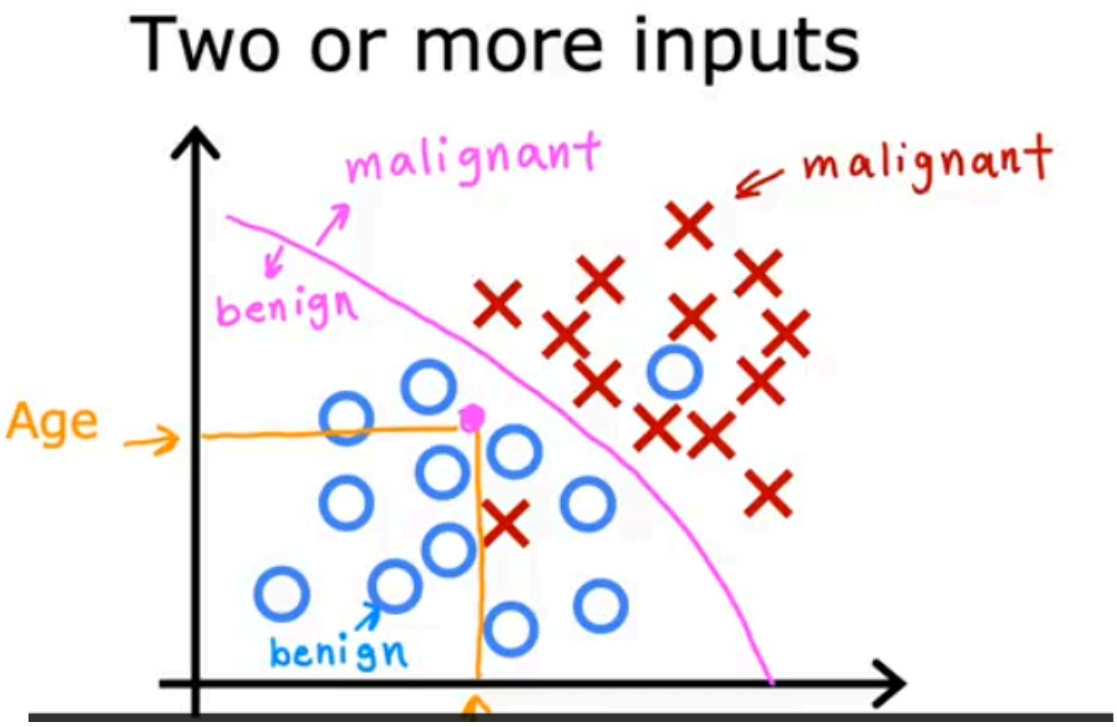
algo that learn the process from given input x and output label y.

> how do we get the desired input and output pair?

- Regression (a type of algorithm learn to deal with numbers)
  Predict a number from infinitely many possible outputs.
- Classification
  Predict a small, finite possible outputs



In the cases we have just covered in class, most of the time we will probably have two ore more inputs

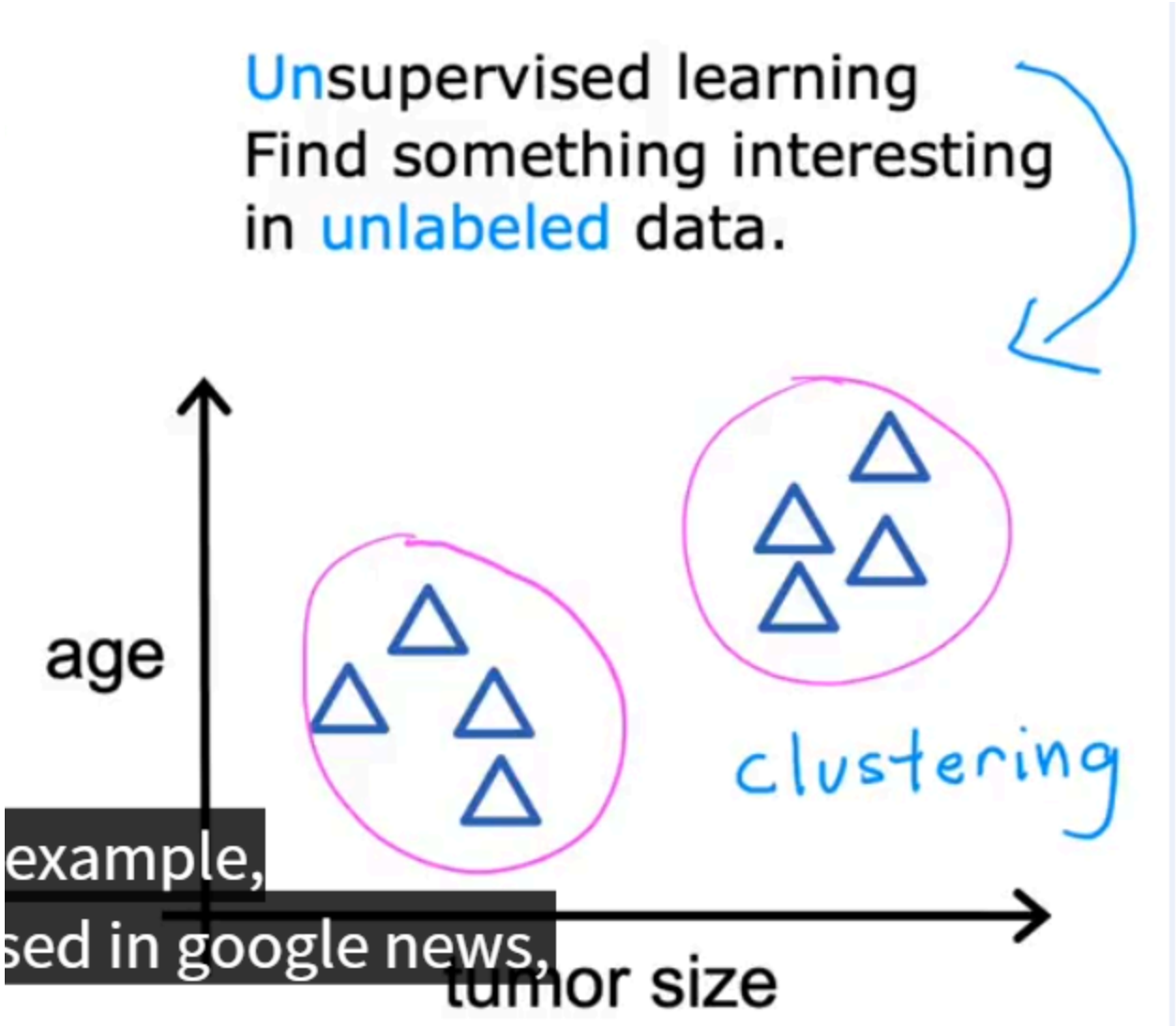


Unsupervised Learning

Contract to supervised learning, we don't give the algorithm the output, we want the algorithm to find interesting patterns.

one common algo of unsupervised learning is

- CLustering



it takes unlabeled data and try to group them into clusters (find the pattern of the data).

other algorithm includes:

- dimentional deduction
- anomolys detection

two kinds of the labs

- optional labs:
  - easy, designed not to write any code
  - need to run
  - take a look in to how machine learning code actually works
- practise lab

- where you write acutual code